Chapter 2 State space, Problem Solving & Representation

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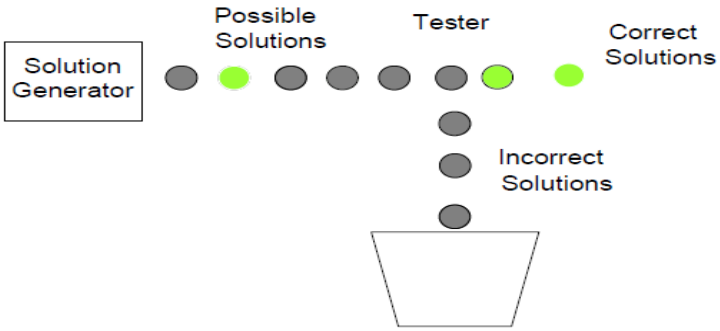
AI as Problem Solving

- Historically many people link intelligence with the capability to solve a problem.
- Problem solving is one of the factors that demonstrate intelligence.

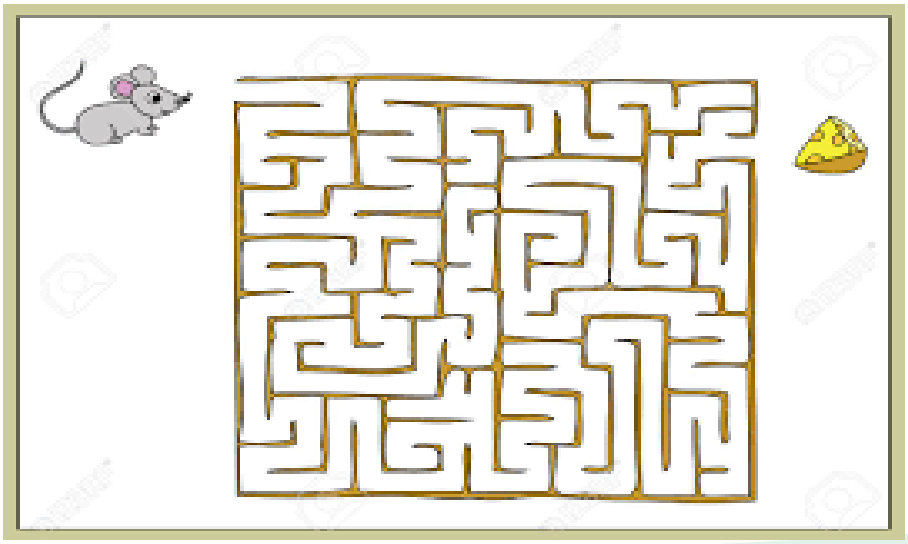
Classical Approach

- The classical approach that is used is "hit and trial" that check for various solutions to that problem.
- It is also known as 'Generate and Test' approach
- We use this approach to solve small problems.



Classical Approach

- Consider the maze searching problem.



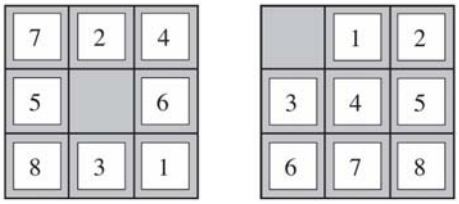
Systematic Problem Solving Approach

- For a simple problem or a smaller problem, we can use the hit-and-trial approach to solve the problem. However, when the problem is complex, then we need a systematic way to solve the problem
- The key to solving a complex problem is its representation.
- Normally, the problems are represented in the form of diagrams (such as graphs or trees).

Problem Solving Process

- **Goal formulation**
  - **Goals organize problem-solving to meet requirements.**
- **Problem formulation**
  - **This creates an abstract model that describes the states and actions required to achieve the goal, organized into a standardized format.**
- **Search for a solution**
  - **This involves searching for solutions until a path to the goal is discovered.**
- **Execution**
  - **Once a solution is found, the search techniques are used to develop a real-world application.**

Example-Standard Formulation



Start State Goal State

- **States:** A state description specifies the location of each of the eight tiles and the blank in one of the nine squares.
- **Initial state:** Any state can be designated as the initial state. Note that any given goal can be reached from exactly half of the possible initial states.
- **Goal state:** The objective of the activity/event.
- **Actions:** The simplest formulation defines the actions as movements of the blank space Left, Right, Up, or Down. Different subsets of these are possible depending on where the blank is.
- **Transition model:** Given a state and action, this returns the resulting state; for example, if we apply Left to the start state in Figure, the resulting state has the 5, and the blank switched.
- **Path cost function:** This checks whether the state matches the goal configuration shown in Figure. (Other goal configurations are possible.)

Tree and Graph Terminologies

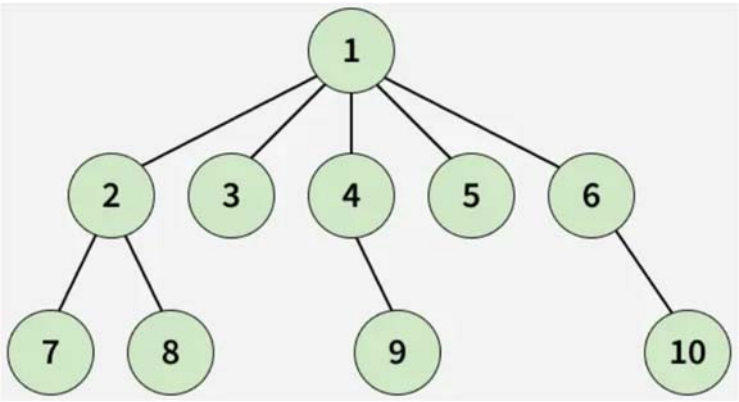
Tree

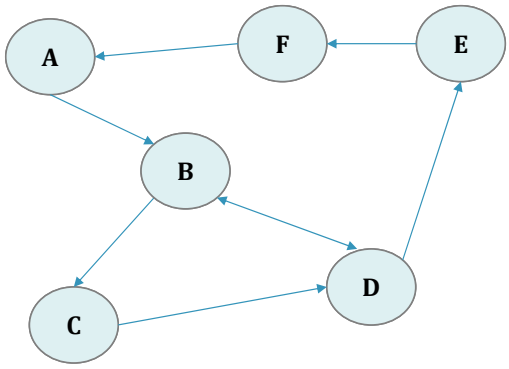
- A tree is a graph in which two nodes have at most one path between them.
- Trees often have roots that is why it is also known as rooted graph.
- In tree cycles are not allowed which means that it is impossible for a path to loop or cycle continuously through a sequence of nodes.

Graph

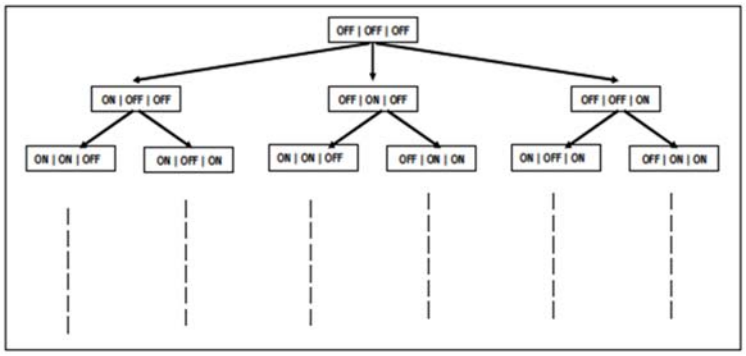
- A graph is a set of nodes and arc/edges that connect them.
- A graph is said to be directed if its arcs contain directionality using arrows.
- If arcs do not contain an arrow, then this graph is known as an undirected graph.

Tree and Graph Terminologies



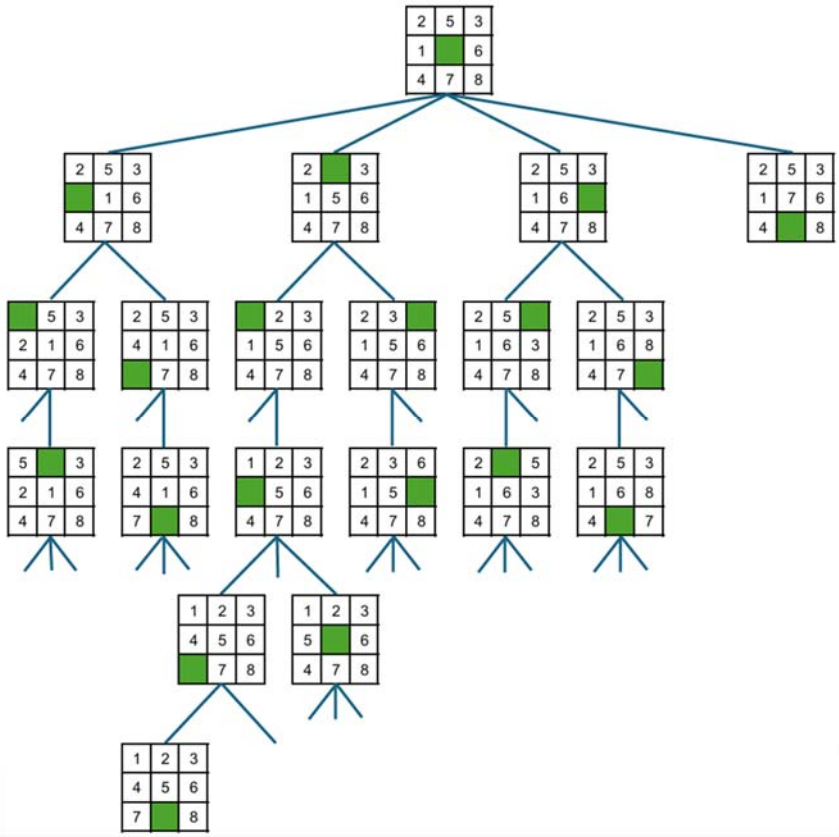


Problem Representation:



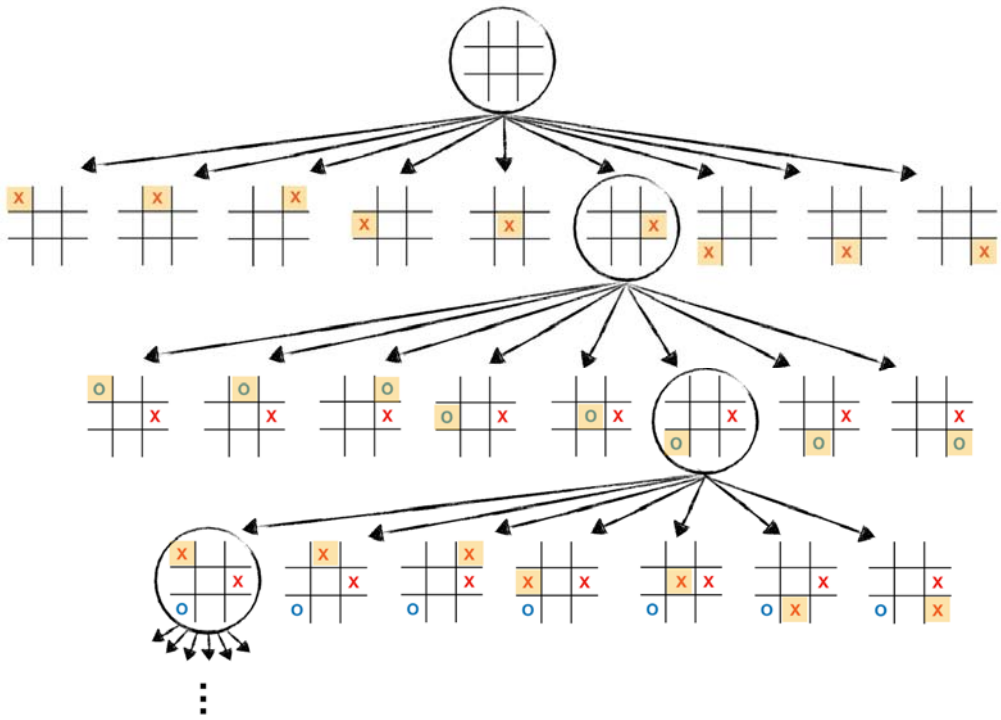
- It shows the problem of switching on the light in a graphical form.
- Each rectangle represents the state of the switchboard.
- OFF | OFF| OFF means that all three switches are OFF.
- OFF| ON | OFF means that the first and the last switch is OFF and the middle one is ON.
- Starting from the state when all the switches are OFF, the person can proceed in any of the three ways by switching either one of the switches ON.
- This brings the person to the next level in the tree. Now, from here, he can explore the other options till he gets to a state where the switch corresponding to the light is ON.

Problem Representation- 8-Puzzle Game:



- States: Position of number 1-8 and space (0)
- Initial state: {{2, 5, 3},{1, 0, 6},{4, 7, 8}}
- Goal state: {{1, 2 ,3},{4, 5, 6},{7, 0, 8}}
- Actions: ACTIONS(Left, Right, Up, Down)
- Transition model: RESULT(0, Left) = {{2, 5, 3},{0, 1, 6},{4, 7, 8}}
- Path cost function:
  - $\text{COST}(0, \text{Left}, \{\{2,5,3\},\{0,1,6\},\{4,7,8\}\}) = 1$
  - $\text{PATH-COST}(0, \text{Left}, \{\{1,2,3\},\{4,5,6\},\{7,0,8\}\}) = n$

Problem Representation-Tic Tac Toe



- States: replace “X” and “O” in the spaces of 3x3 matrices
- Initial state: {{X, – , –},{–, O, –}, {O,–,–}}
- Goal state: {{X, O, X},{–, O, –}, {O,–,X}}
  - Winner X, Winner O, Draw/Tie
- Actions: Place(*r*, *c*)
- Transition model:
  - Input: current state+ Place(r,c)+ current player
  - Output: new state with new turn to the opposite player
- Path cost function:
  - +1 (win), -1 (loss), 0 (tie)

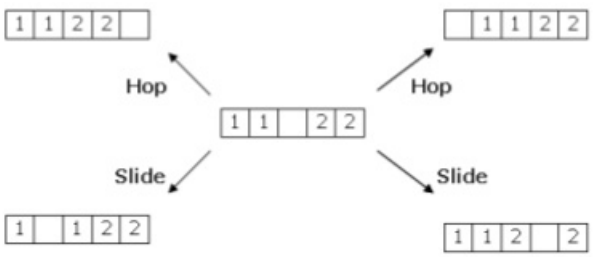
X		
	O	
O		

Initial state

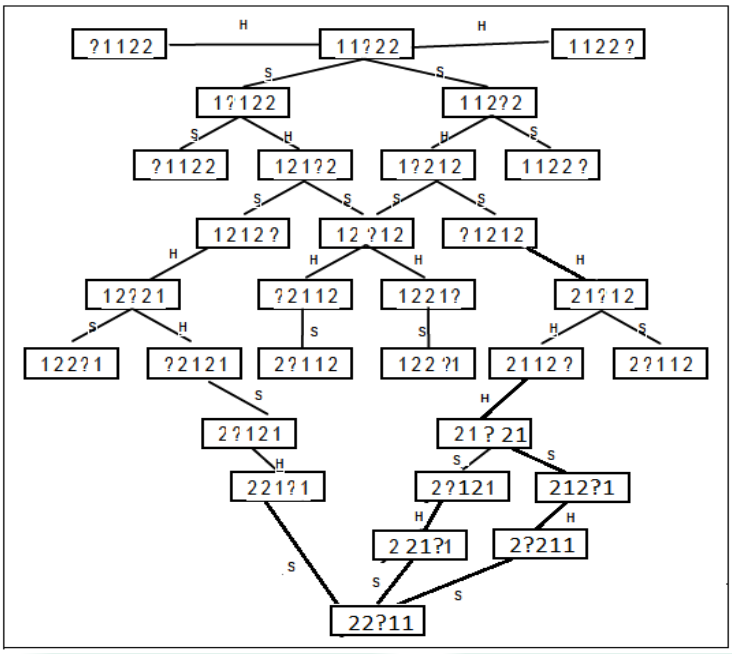
X	O	X
	O	
O		X

Goal state

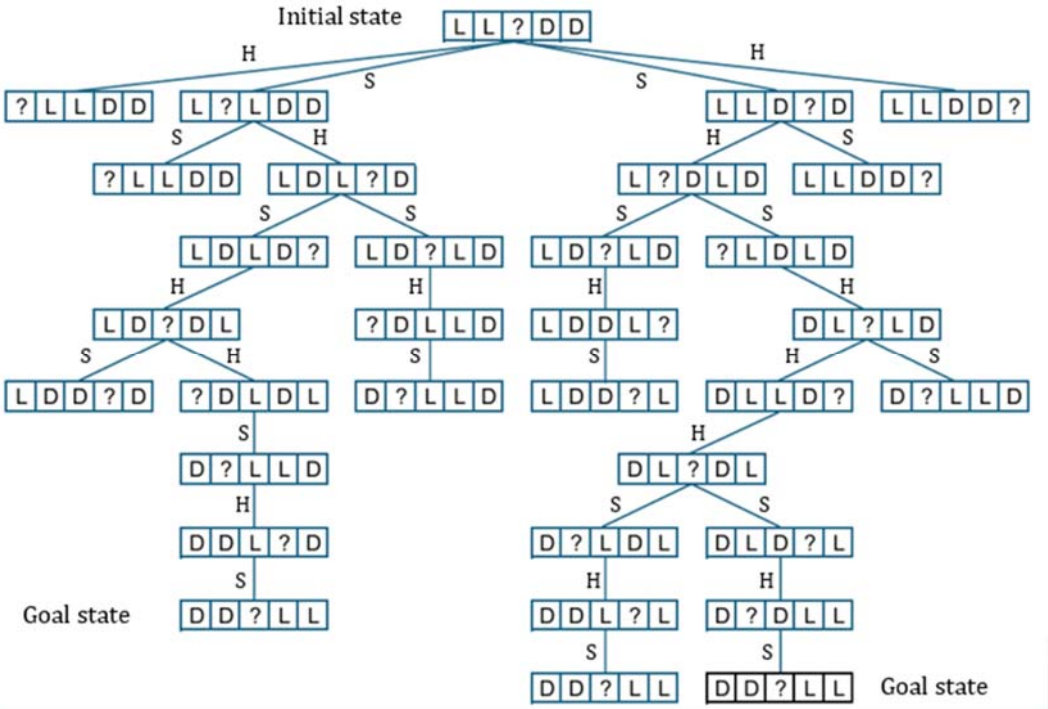
Problem Representation- The Two-One Problem



All possibilities of first state

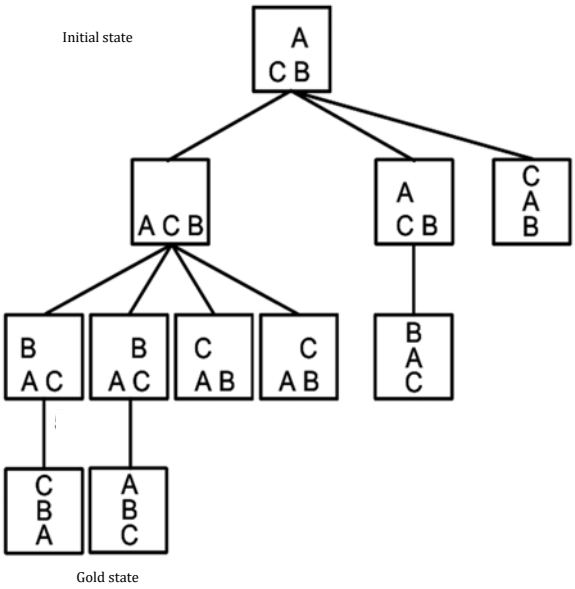


Problem Representation-Frog Puzzle



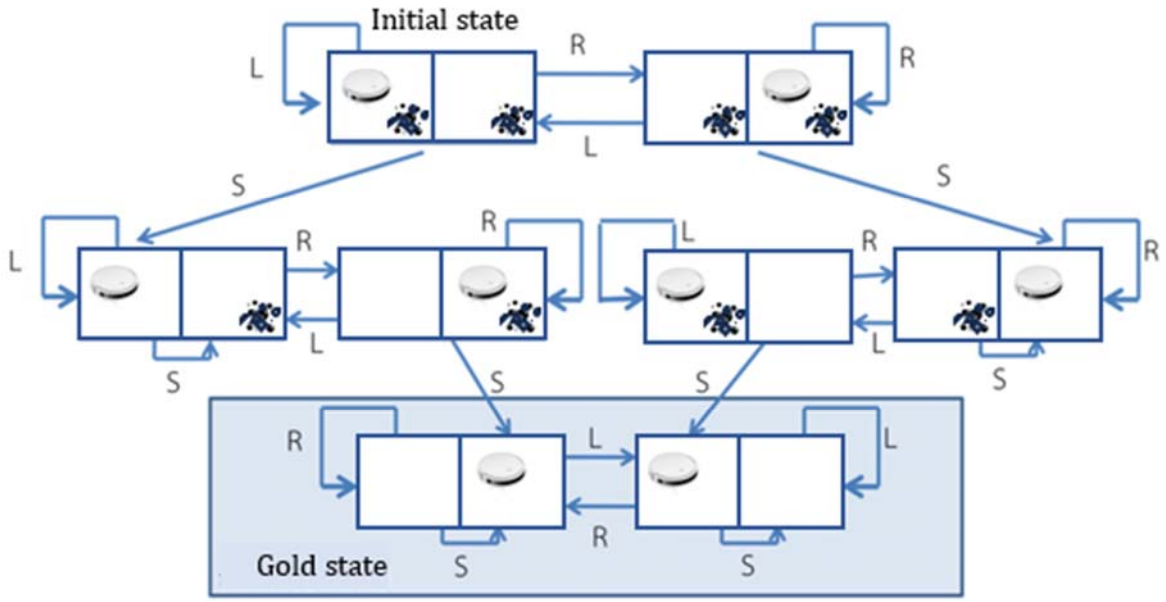
- States: replace
- Initial state:
- Goal state:
- Actions:
- Transition model:
- Path cost function:

Block world



- States: replace
- Initial state:
- Goal state:
- Actions:
- Transition model:
- Path cost function:

Vacuum world



- States: replace
- Initial state:
- Goal state:
- Actions:
- Transition model:
- Path cost function:

Problem Representation-The Traveling Salesperson

- Suppose a salesperson has five cities to visit and then must return home. The goal of the problem is to find the shortest path for the salesperson to travel, visiting each city, and them returning to the starting city (A)

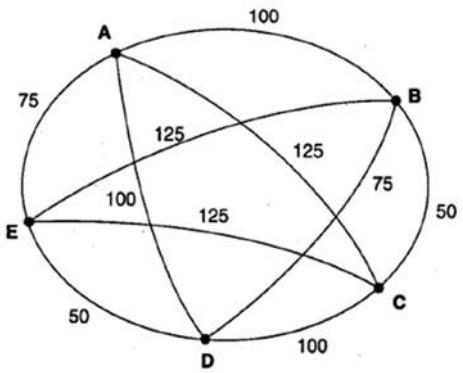
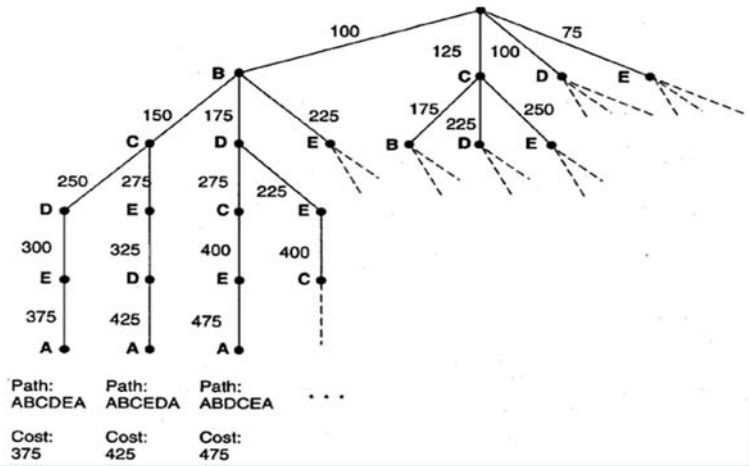
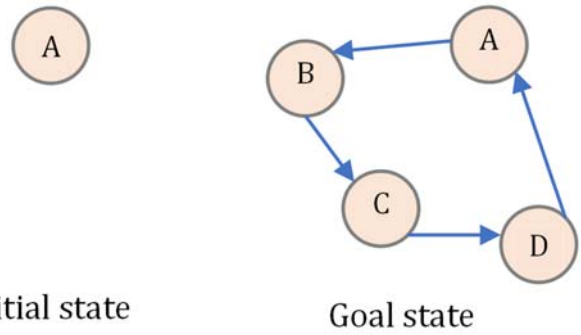


Figure An instance of the traveling salesperson problem.



Travel Salesperson Problem (TSP)



- States: replace
- Initial state:
- Goal state:
- Actions:
- Transition model:
- Path cost function: